

The analytic bridge: Cauchy coefficients in one and several variables (Taylor-tree Stages 6–7)

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Abstract

Units 257–265 remove the qualification under which the Taylor-tree programme was frozen at unit 256: the theorem `thm:TaylorTree` had been formalised for phase and amplitude given by coefficient families with $\sum_{\gamma} |c_{\gamma}| b^{|\gamma|} < \infty$, but not under the paper’s own hypothesis (holomorphic extension to a polydisc D_R , $R > b$). Stage 6 (two units) does the one-variable case with Mathlib’s `cauchyPowerSeries`. Stage 7 (six units) builds the several-variable Cauchy theory that Mathlib lacks, from a normalised circle operator iterated along the coordinates, and lands Headline XXXII: for real ξ, η on $(0, b]^d$ with holomorphic extensions F_{ξ}, F_{η} to $\{|z_i| < R\}$, $R > b$, the real parts of the Cauchy coefficients at any radius $r \in (b, R)$ are summable families representing ξ, η on the box, the family integral is the original $Z(\beta, n; \xi, \eta)$, and the Taylor-tree conclusion holds — in every dimension. Reviews v24 and v25: qualified passes with no mathematical defect. Remaining gap to the paper’s formulation: the identification $c_{\gamma} = \partial^{\gamma} F(0)/\gamma!$. No sorry, no additional axiom; 9140 jobs; pin 45b0943.

1 Stage 6: one variable (units 257–259)

Mathlib’s API sufficed in one unit. For f differentiable on the closed disc $|z| \leq r$, `DifferentiableOn.hasFPowerSeriesOnBall` gives a power series on the open disc whose coefficients are `cauchyPowerSeries f 0 r; norm_cauchyPowerSeries_le` with `FormalMultilinearSeries.norm_apply_eq_norm_coef` gives $|c_n| \leq M_r r^{-n}$; `HasFPowerSeriesOnBall.hasSum` with `apply_eq_pow_smul_coef` gives $\sum c_n z^n = f(z)$; and `AnalyticAt.hasFPowerSeriesAt` (the series $\sum f^{(n)}(0)/n! z^n$) with `HasFPowerSeriesAt.eq_formalMultilinearSeries` gives $c_n = f^{(n)}(0)/n!$. Real parts are taken by `HasSum.mapL Complex.reCLM`, which needs no reality hypothesis on f : the family of real parts represents $\operatorname{Re} f$ on the segment, and equals f there when f is real. Unit 259 (review v24 should-fixes) added the paper-facing form: from $0 < b < R$ and holomorphy on the open disc, choose $r \in (b, R)$; the family integral on $(0, b]$ is transported to the original integral by `setIntegral_congr_fun` with the representation.

2 Stage 7: several variables (units 260–265)

Mathlib has no several-variable Cauchy estimates or polydisc power series. Astra #31 chose route R2: induct on the dimension with a *normalised circle operator*

$$A_r(g) = \frac{1}{2\pi i} \oint_{|w|=r} \frac{g(w)}{w} dw,$$

applied to *slices of F*, never to coefficient functions, with the gate “quantitative two-coordinate reconstruction” after unit three. The gate was passed at unit 263 and the tranche closed in six units against a cap of ten.

Circle operator (u260). The bound $\|A_r g\| \leq \sup_{|w|=r} \|g\|$ comes from `circleIntegral.norm_integral_le_of_norm_le_const`, since $|2\pi i|^{-1} \cdot 2\pi r \cdot r^{-1} = 1$; the Cauchy formula $A_r(w \mapsto (1 - z/w)^{-1} g(w)) = g(z)$ for $|z| < r$ and g differentiable on a larger ball, from `DiffContOnCl.circleIntegral_sub_inv_smul` after rewriting $w^{-1}(1 - z/w)^{-1} = (w - z)^{-1}$; coefficient `HasSum` $\sum_n A_r(w \mapsto w^{-n} g(w)) z^n = g(z)$ from the geometric expansion of the kernel and dominated interchange (`intervalIntegral.hasSum_integral_of_dominated_convergence` on the parametrisation); continuity of $z \mapsto A_r(G(\cdot, z))$ from `intervalIntegral.continuous_parametric_intervalIntegral_of_continuous`.

Iterated operator and iterated Cauchy formula (u261). $A_r^{[d+1]}(G) = A_r(w_0 \mapsto A_r^{[d]}(w' \mapsto G(\text{cons } w_0 w')))$, recursive along `Fin.cons`. The hypothesis `SliceHolo d r F` is recursive: continuity on the closed polydisc plus, for each w_0 on the circle, `SliceHolo` of the slice $F(w_0, \cdot)$, plus one-variable holomorphy of $w_0 \mapsto A_r^{[d]}$ -images; it is discharged from `DifferentiableOn` on the open polydisc by `differentiable_pi` and `differentiable_apply`. The iterated Cauchy formula $A_r^{[d]}(w \mapsto F(w) \prod_i (1 - z_i/w_i)^{-1}) = F(z)$ is the induction with the one-variable formula applied to $w_0 \mapsto F(w_0, z')$.

Parametric continuity and series interchange (u262). On sets: `continuousOn_iff_continuous_domRestrict` converts the joint continuity of G on $\text{circle} \times S$ to a genuine continuity, and the iterated version recurses. Interchange $A_r(\sum_n G_n) = \sum_n A_r(G_n)$ under a summable sup-majorant; the product geometric series $\sum_\gamma \prod_i (|z_i|/r)^{\gamma_i}$ as `summable_prodGeom`.

Cauchy coefficients and the gate (u263). $c_\gamma = A_r^{[d]}(F \prod_i w_i^{-\gamma_i})$; $|c_\gamma| \leq M r^{-|\gamma|}$ needs only a bound on the torus (Lean’s totalised contour integrals ask for no holomorphy there). Reconstruction: extract the first coordinate with the one-variable coefficient `HasSum`, expand the tail slice by the induction hypothesis *on the circle*, interchange A_r with the tail series under the product-geometric majorant, and regroup the double series along `Fin.cons` with `HasSum.sigma` and `Fin.consEquiv`. Consequence: $\sum_\gamma |c_\gamma| b^{|\gamma|} \leq M(1 - b/r)^{-d}$ for $b < r$.

Headline XXXII (u264–265). Boundedness on the closed polydisc of radius r comes from compactness (`isCompact_univ_pi`) inside the open polydisc of radius R ; the real-part family `polyRealCoeff` is `AbsSummableAt b` and represents $\text{Re } F$ on the box via `HasSum.mapL Complex.reCLM`; the family integral is transported to the original integral $Z(\beta, n; \xi, \eta)$ pointwise on the box. The wrapper `thm_TaylorTree_coeffFamily` then gives `thm_TaylorTree_analytic`; the primed form takes only $b < R$.

3 Reviews and numerics

Review v24 (units 256–258) and v25 (units 260–264) are qualified passes with no mathematical defect. Review v25 asked for: “applies under the paper’s hypothesis” rather than “exactly” (the formal matching assumption is real-part agreement $\text{Re } F_\xi = \xi$ on the box, weaker than a genuine extension); the residual task stated with the complex/real distinction ($c_\gamma = \partial^\gamma F(0)/\gamma!$, Re of it

for the family, and the linkage to derivatives of the real ξ at 0 needing the genuine extension); an $\exists r$ corollary; and the distinction between the torus bound used for the estimate and the closed-polydisc bound used for the reconstruction. All applied in unit 265. Numerically, at $d = 2$ with $F = e^{z_0+2z_1} + z_0 z_1^2$ and $r = 0.7$, the iterated contour coefficients agree with the Taylor coefficients to 2×10^{-16} and the truncated reconstruction on the open polydisc to 4×10^{-15} .

4 Lessons

- **Recurse on slices of the function, not on coefficient functions.** Coefficient functions $w_0 \mapsto c_{\gamma'}(F(w_0, \cdot))$ would need parametric holomorphy of contour integrals; slices need only continuity, which the parametric-integral lemmas give directly.
- **Normalise the operator.** With $(2\pi i)^{-1} \oint w^{-1}$ the operator norm is exactly one, so torus bounds pass through the recursion without radius bookkeeping.
- **circleIntegral is a definition, not an interface.** Its unfolding $\int_0^{2\pi} \text{deriv}(\text{circleMap } c R) \theta \bullet f(\dots)$ appears whenever the integrand is a lambda that Mathlib's lemmas do not recognise; fix the integrand with a `funext` equation and `circleIntegral.integral_smul` before applying anything. Also `circleMap_ne_center` takes $R \neq 0$ with θ implicit, and `Fin.cons w w' i` needs the type ascription `(Fin.cons w w' : Fin (d+1) -> C) i` to elaborate.
- **Apply DifferentiableOn.diffContOnC1_ball to the larger ball**, then restrict; the closure of the smaller ball is where continuity is needed.
- **Renamed lemmas on this pin:** `tendsto_finsetSum`, `integral_finsetSum`, `continuous_finsetProd`, `continuousOn_finsetProd`, `Metric.mem_eball`, `continuousOn_iff_continuous_domRestrict`.
- **Statement-extractor artefact.** The review excerpts are produced by a Python extractor that sometimes duplicates or truncates a declaration; reviewers then report “duplicate declaration”. The source always has one; say so in the prompt.
- **Hand-off before extension.** Freezing at unit 256 with an explicit gap statement made the bridge a separately reviewable tranche with its own gate and cap, rather than an open-ended extension.